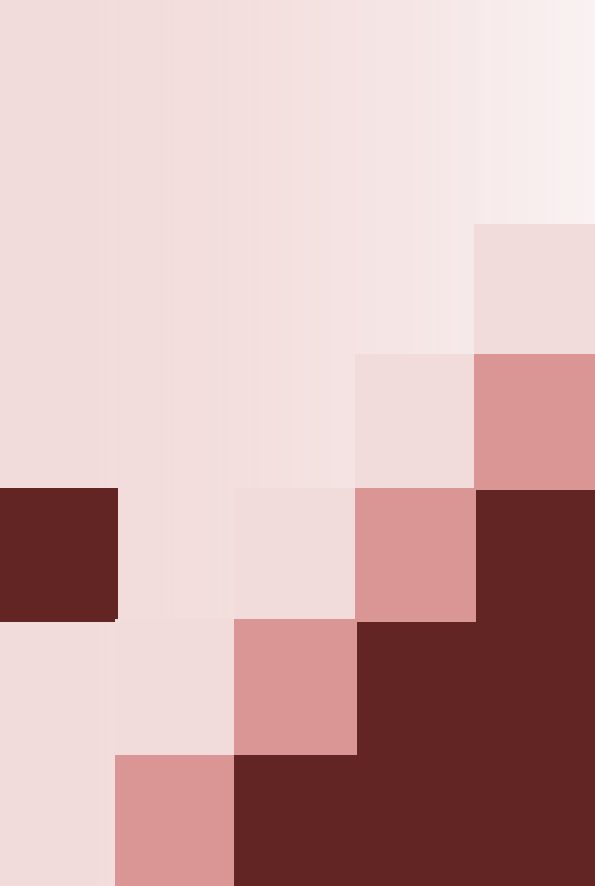




Cloud Computing Fundamentals

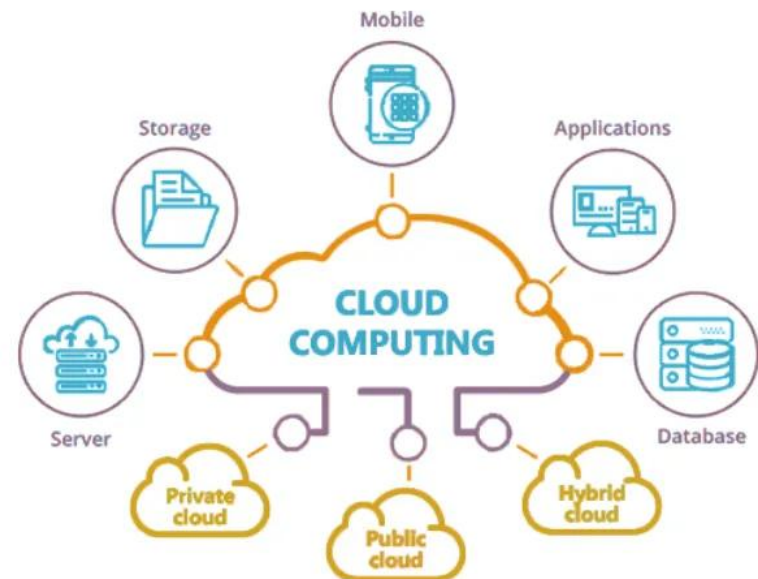
Rajeev Shorey

IIIT Surat

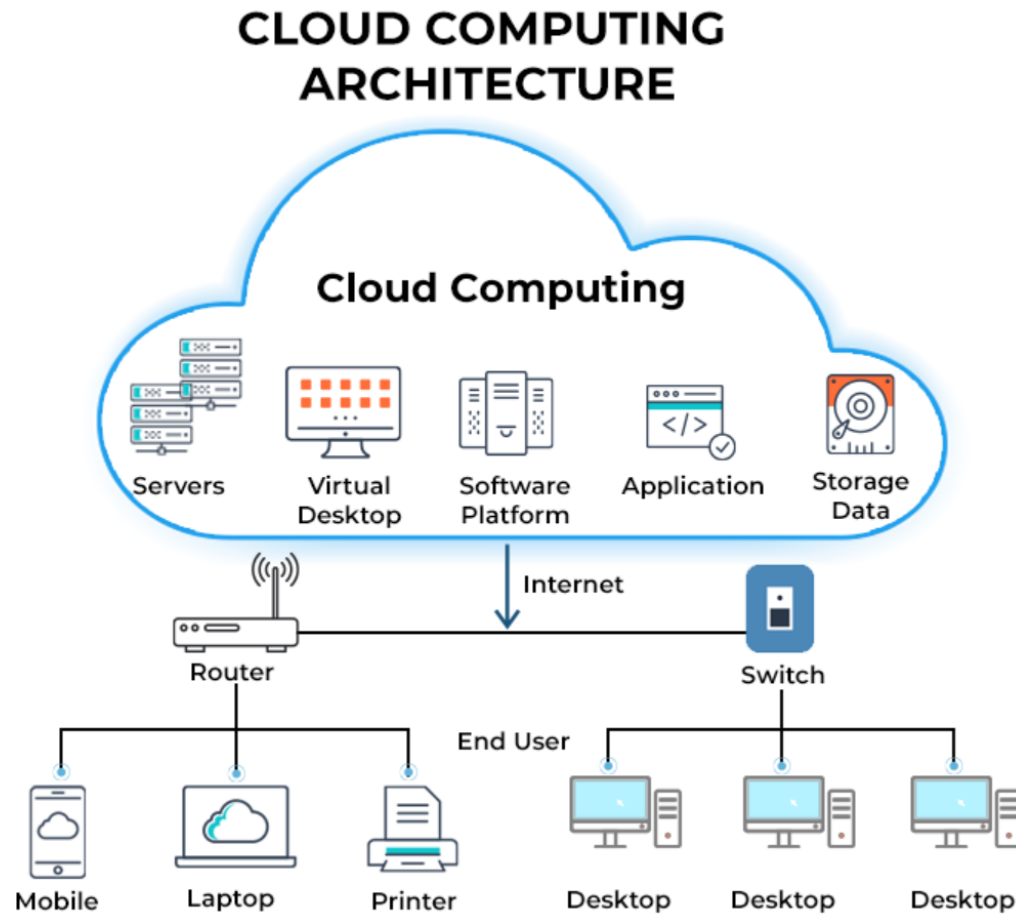
31 July 2025

Cloud Computing Definition

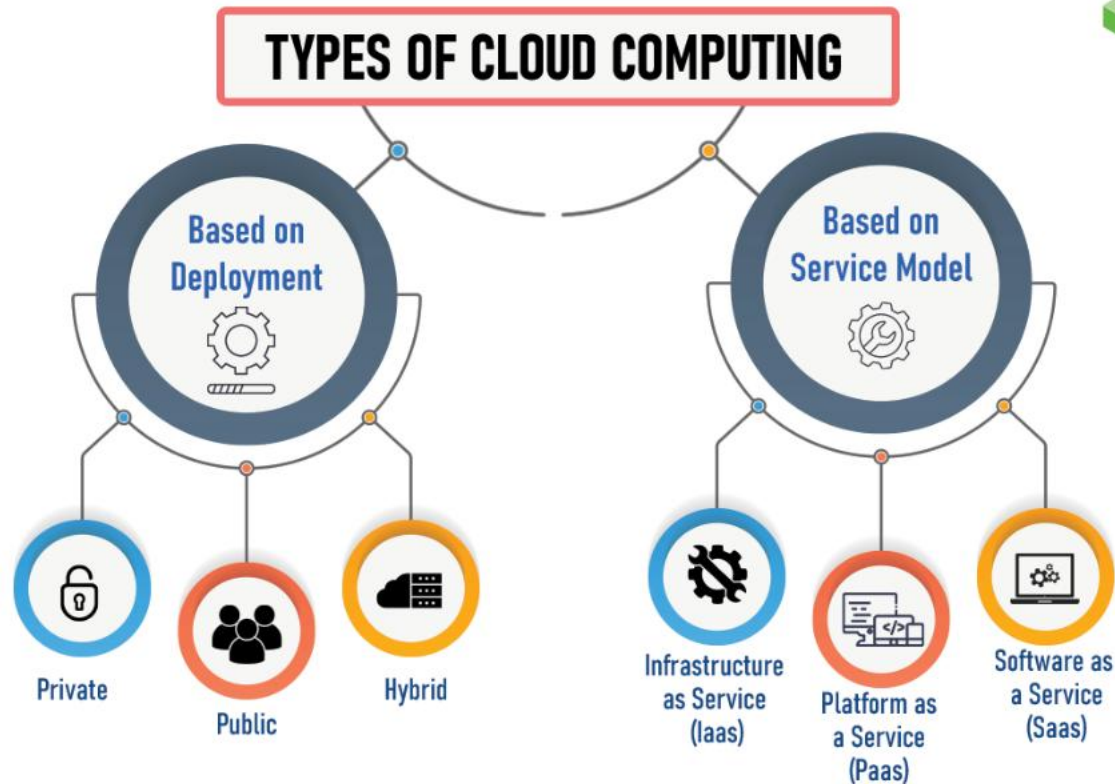
Cloud computing is the on-demand availability of computing resources (such as storage and infrastructure), as services over the internet. It eliminates the need for individuals and businesses to self-manage physical resources themselves, and only pay for what they use.



Cloud Computing Architecture



Types of Cloud Computing



Cloud infrastructure management

■ Management models:

- **Public clouds**

- Utility model
- Shared hardware
- No control of hardware,
- Self-managed (e.g., AWS, Azure, GCE)

- **Private clouds:**

- More isolated (secure?),
- Federal compliance friendly
- Customizable hardware and hardware sharing
- Example: OpenStack, an open source platform for virtualized clouds

Common cloud computing models

■ Service models:

- **IaaS:** Infrastructure as a Service
 - e.g., Amazon EC2, Microsoft Azure, Google Compute Engine
- **PaaS:** Platform as a Service
 - e.g., Google AppEngine, Heroku, Apache Stratos
- **SaaS:** Software as a Service
 - Microsoft Office365, Amazon DynamoDB, gmail

Cloud Computing Benefits

Cloud Computing Benefits



01 FASTER AND
FLEXIBLE
SCALABILITY

02 REDUCED IT
COST

03 IMPROVED BUSINESS
INTERACTION AND
COLLABORATION

04 ENHANCED
BUSINESS VALUE

05 EASY DATA
BACKUP AND
RESTORE

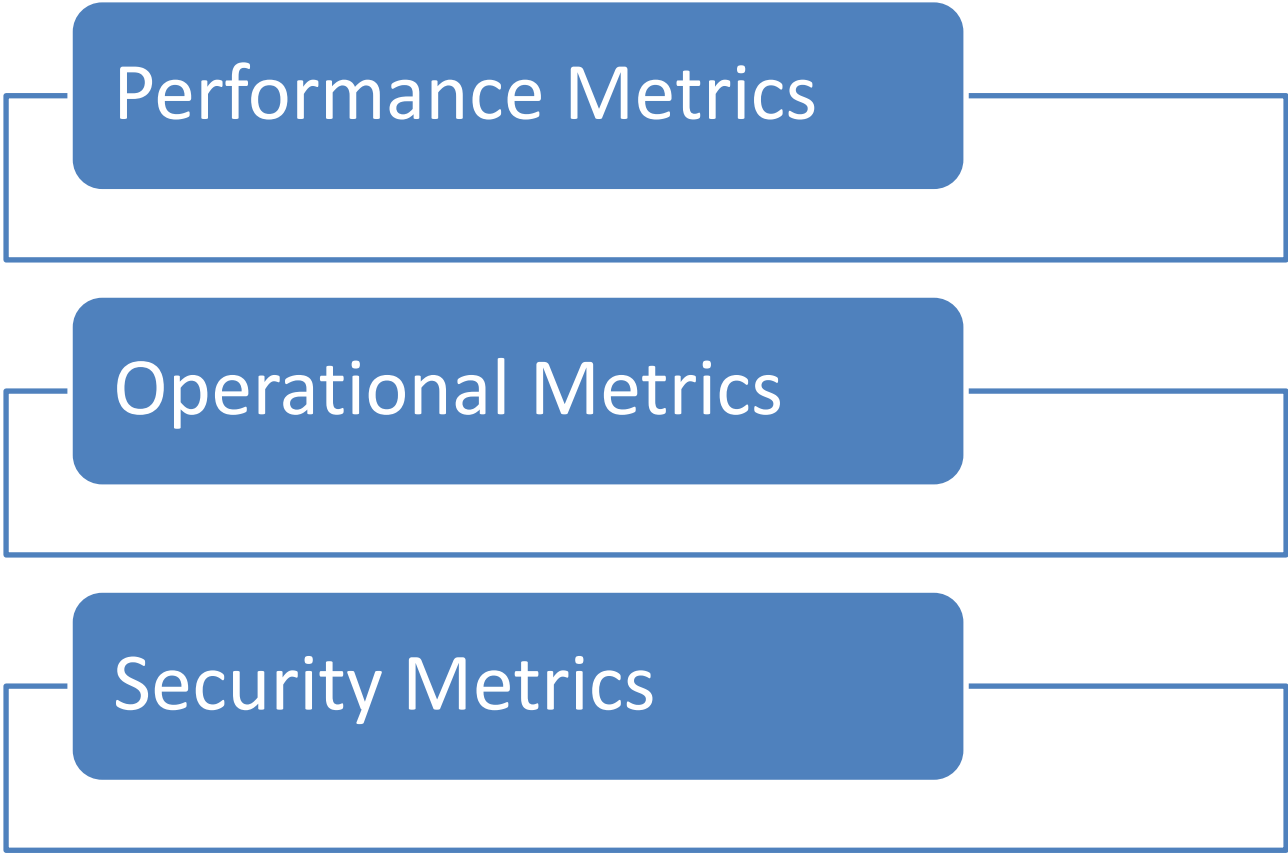
06 COMPETITIVE
EDGE AND
SUSTAINABILITY

07 FLEXIBLE
STORAGE

08 MOBILITY AND
REMOTE WORK

09 BUSINESS
CONTINUITY WITH
ANYWHERE ACCESS

Cloud Computing: Performance Metrics



Performance Metrics

Operational Metrics

Security Metrics

Cloud Computing: Performance Metrics

1. Availability

2. CPU utilization

3. Memory

4. Disk usage and I/O

5. Load average

6. Latency

7. Network bandwidth

8. Error rate

9. Requests per minute

10. Mean time to repair (MTTR)

Cloud Computing: Performance Metrics

Energy Usage Effectiveness

Carbon Intensity Per Task

Reliability

Fault Tolerance

Scalability

Virtualization: key to cloud boom

- Physical machine virtualization
 - Full virtualization: masking as physical machine to guest
 - Para-virtualization: Semi-transparent to guest
- Virtualization goes well beyond CPU or physical machines
 - Virtualized I/O (Elastic Block Storage)
 - Virtual private networks
 - Containers (e.g., AWS or Google Container Service)
 - Virtual switches/routers with software-defined networking (SDN)
 - Network function virtualization (NFV) forwarding-plane services

Cloud Pricing Models

- Market-driven: variety of pricing models for each of IaaS, PaaS, SaaS service models
- Let's ask some questions:
 - How much does an hour of computing on a commodity core cost today?
 - How much does transferring 1GB of data over a wide-area network cost today?
 - How much does 1GB of storage for a month cost today?

Let's find out what the market looks like today!

Elastic Compute Cloud (EC2) Pricing

	vCPU	ECU	Memory (GiB)	Instance Storage (GB)	Linux/UNIX Usage
General Purpose - Current Generation					
t2.nano	1	Variable	0.5	EBS Only	\$0.0065 per Hour
t2.micro	1	Variable	1	EBS Only	\$0.013 per Hour
t2.small	1	Variable	2	EBS Only	\$0.026 per Hour
t2.medium	2	Variable	4	EBS Only	\$0.052 per Hour
t2.large	2	Variable	8	EBS Only	\$0.104 per Hour
m4.large	2	6.5	8	EBS Only	\$0.12 per Hour
m4.xlarge	4	13	16	EBS Only	\$0.239 per Hour
m4.2xlarge	8	26	32	EBS Only	\$0.479 per Hour
m4.4xlarge	16	53.5	64	EBS Only	\$0.958 per Hour

Do these numbers match our intuition based on market prices for computers and peripherals?

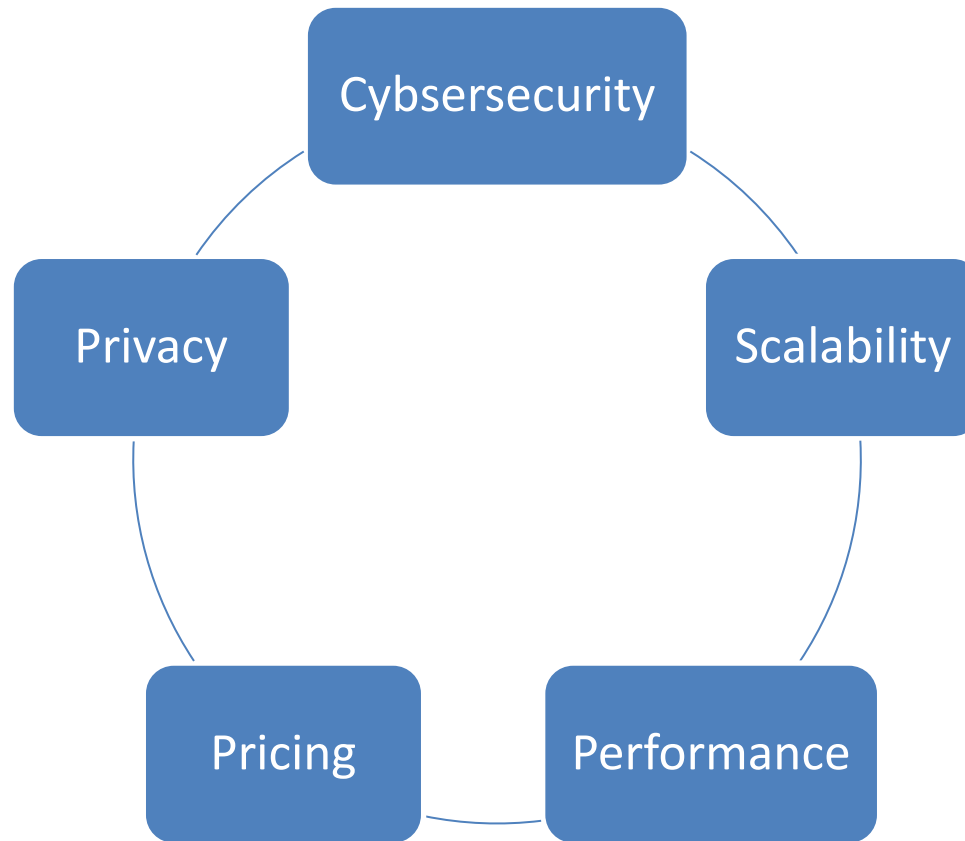
Google Cloud compute pricing

Machine type	Virtual CPUs	Memory	GCEU ¹	Lowest price ² (USD) per hour with full sustained usage	Typical price ³ (USD) per hour	Full price ⁴ (USD) per hour without sustained use	Preemptible price ⁵ (USD) per hour
n1-standard-1	1	3.75GB	2.75	\$0.035	\$0.038	\$0.050	\$0.015
n1-standard-2	2	7.5GB	5.50	\$0.070	\$0.076	\$0.100	\$0.030
n1-standard-4	4	15GB	11	\$0.140	\$0.152	\$0.200	\$0.060
n1-standard-8	8	30GB	22	\$0.280	\$0.304	\$0.400	\$0.120
n1-standard-16	16	60GB	44	\$0.560	\$0.608	\$0.800	\$0.240
n1-standard-32 ⁶	32	120GB	88	\$1.120	\$1.216	\$1.600	\$0.480

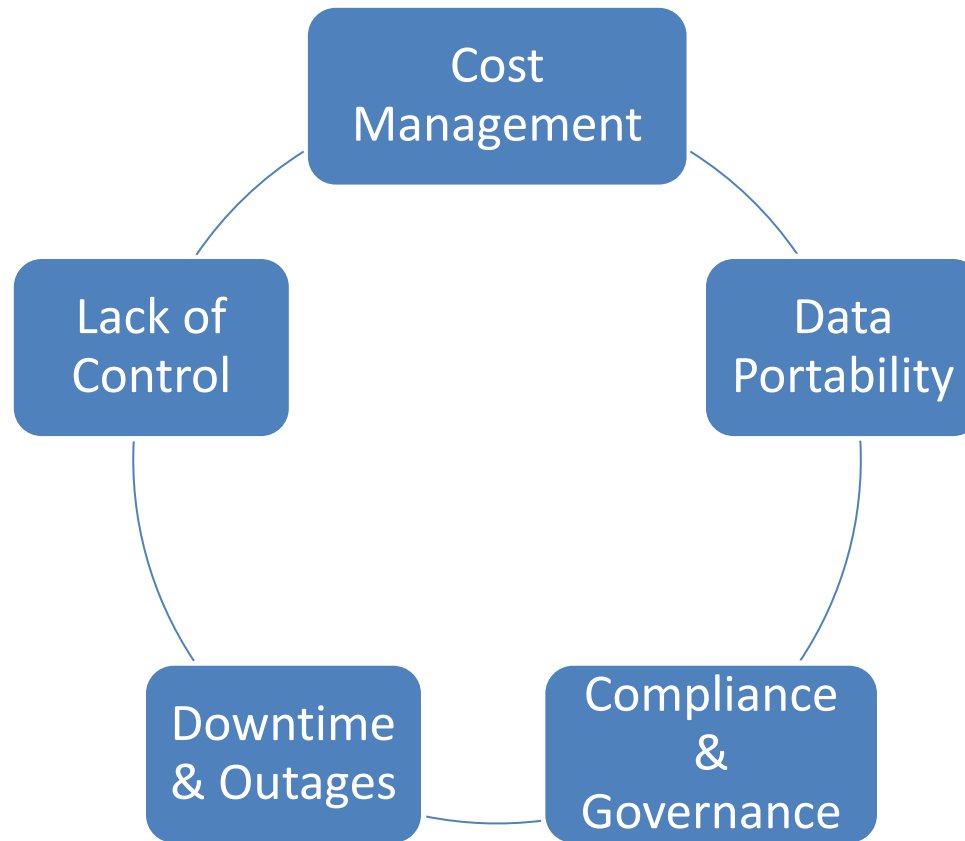
Spot pricing: market inside a cloud

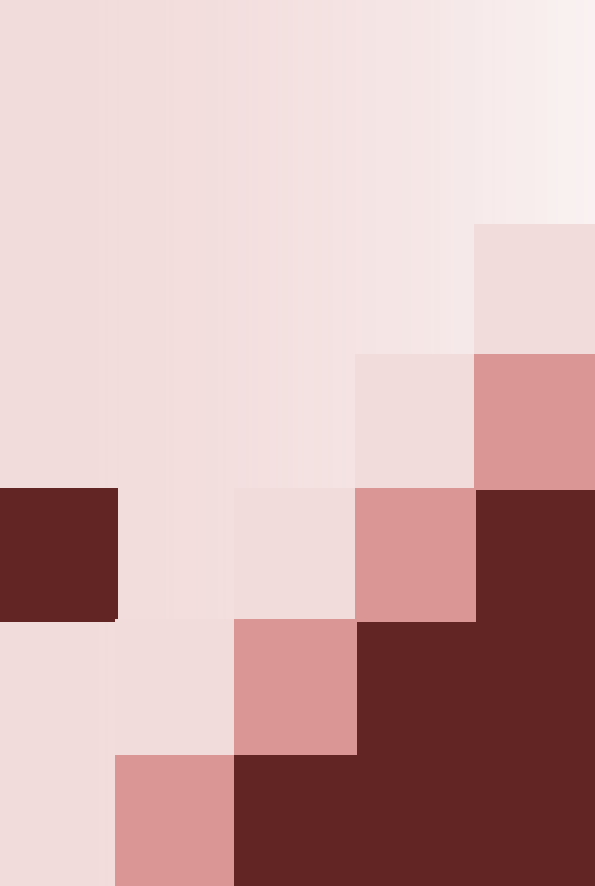
	Linux/UNIX Usage	Windows Usage
General Purpose - Current Generation		
m3.medium	\$0.0098 per Hour	\$0.0591 per Hour
m3.large	\$0.0196 per Hour	\$0.1171 per Hour
m3.xlarge	\$0.0379 per Hour	\$0.1382 per Hour
m3.2xlarge	\$0.075 per Hour	\$0.2752 per Hour
m4.large	\$0.027 per Hour	\$0.1391 per Hour
m4.xlarge	\$0.0468 per Hour	\$0.2785 per Hour
m4.2xlarge	\$0.0974 per Hour	\$0.5557 per Hour
m4.4xlarge	\$0.201 per Hour	\$1.1156 per Hour
m4.10xlarge	\$0.5948 per Hour	\$2.9799 per Hour

Key Challenges in Cloud Computing



Additional Challenges in Cloud Computing





Edge Computing

Edge Computing

EDGE COMPUTING

